SIGMACAMP · ELECTROCHEMISTRY LAB · SESSION 1

Fruit Battery

Experiment Packet

Audience: instructors and teaching assistants

Purpose: Bench instructions for instructors and TAs

Session hook: Familiar objects make electricity

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

SESSION OVERVIEW & PLAN

Session 1 — Build a Fruit Battery

Hook: Familiar objects make electricity.

Learning outcomes

By the end of this session, students should be able to:

- Build a simple electrochemical cell from fruit and metal electrodes
- Explain oxidation and reduction in a galvanic cell
- Identify anode and cathode and describe electron vs. ion flow
- Measure voltage with a multimeter
- Explain why cells in series add voltage
- Describe why batteries eventually die (internal resistance, reactant depletion)

Session plan (90 min)

Time	Activity	Notes
0–10 min	Hook: measure one lemon cell; ask whether a lemon can make electricity	See lecture.md — Opening hook
10–25 min	Concept: zinc oxidation, copper cathode, acid electrolyte, electron flow	See lecture.md
25–45 min	Build one cell; measure voltage	See experiment.md — Part 1
45–60 min	Series battery challenge	See experiment.md — Part 2
60–75 min	Power LED, buzzer, or clock; discuss V, I, and internal resistance	See experiment.md — Part 3
75–90 min	Side experiments: fruits, metals, pH, salt, parallel; wrap-up + preview Session 2	See experiment.md — Extensions

Key reactions

Anode: $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

Cathode: $2H^+(aq) + 2e^- \rightarrow H_2(g)$

Overall: $Zn(s) + 2H^+(aq) \rightarrow Zn^{2+}(aq) + H_2(g)$

Files in this folder

File	Purpose
lecture.md	~20 min concept block
experiment.md	Hands-on procedures and data
materials.md	Session checklist
preparation.md	Pre-class setup

Instructor reminders

- Roll lemons/limes first to break internal membranes
- Cells in series add voltage ~linearly; current stays limited
- Galvanized nail = steel + zinc coating (zinc is the active metal)
- pH near cathode may become less acidic (protons consumed)

Status

- [ ] Lecture outline finalized
- [ ] Experiment procedure tested
- [ ] Materials procured
- [ ] Session 2 $CuSO_4$ crystals on hand for next day

PREPARATION

Session 1 — Preparation

Before class (instructor)

- [ ] Procure fresh fruit (firm, not dried out)
- [ ] Test one cell: expected ~0.8–1.0 V for lemon + Zn + Cu
- [ ] Verify multimeters have fresh batteries and work on DC voltage
- [ ] Pre-cut copper strips if students struggle with stiff wire
- [ ] Prepare waste collection for fruit and electrodes

Day-of setup (15 min before)

- [ ] One station per group: fruit, electrodes, wires, multimeter, load device
- [ ] Optional extension materials at a side table

After class

- [ ] Dispose of fruit waste appropriately
- [ ] Rinse and dry electrodes if reusing
- [ ] Confirm Session 2 $CuSO_4 \cdot 5H_2O$ is on site for tomorrow's plating bath

Open questions / notes

- Typical voltage achieved with our materials:
- Best fruit type in pilot:
- LED lighting requires ___ cells in series:

MATERIALS CHECKLIST

Session 1 — Materials

Per group (3–4 students)

- [ ] 3–6 lemons, limes, potatoes, apples, or oranges
- [ ] Galvanized nails or zinc strips (2+)
- [ ] Copper strips, copper wire, or copper coins/pennies (2+)
- [ ] Alligator clip wires (4–6)
- [ ] Multimeter
- [ ] LED, small buzzer, or low-power digital clock
- [ ] pH paper or pH indicator strips
- [ ] Paper towels

Optional (extensions)

- [ ] Graphite pencil leads
- [ ] Aluminum foil
- [ ] Steel nails
- [ ] Magnesium ribbon
- [ ] Table salt (small amount for salt-addition test)

Instructor / room

- [ ] Waste bin for used fruit (not food waste)
- [ ] Hand-washing access
- [ ] Demo multimeter on projector or large display (optional)

Quantities to estimate

Item	Groups	Total
Fruit (minimum)	× 4 each	
Multimeters	1 per group	
9 V batteries	—	Not needed this session

EXPERIMENT PROCEDURE

Session 1 — Experiment: Fruit Battery

Main experiment: Lemon/lime/potato battery — generating electricity from chemistry.

> **Instructors:** expected voltages, LED failure modes, and extension interpretation — instructor.md.

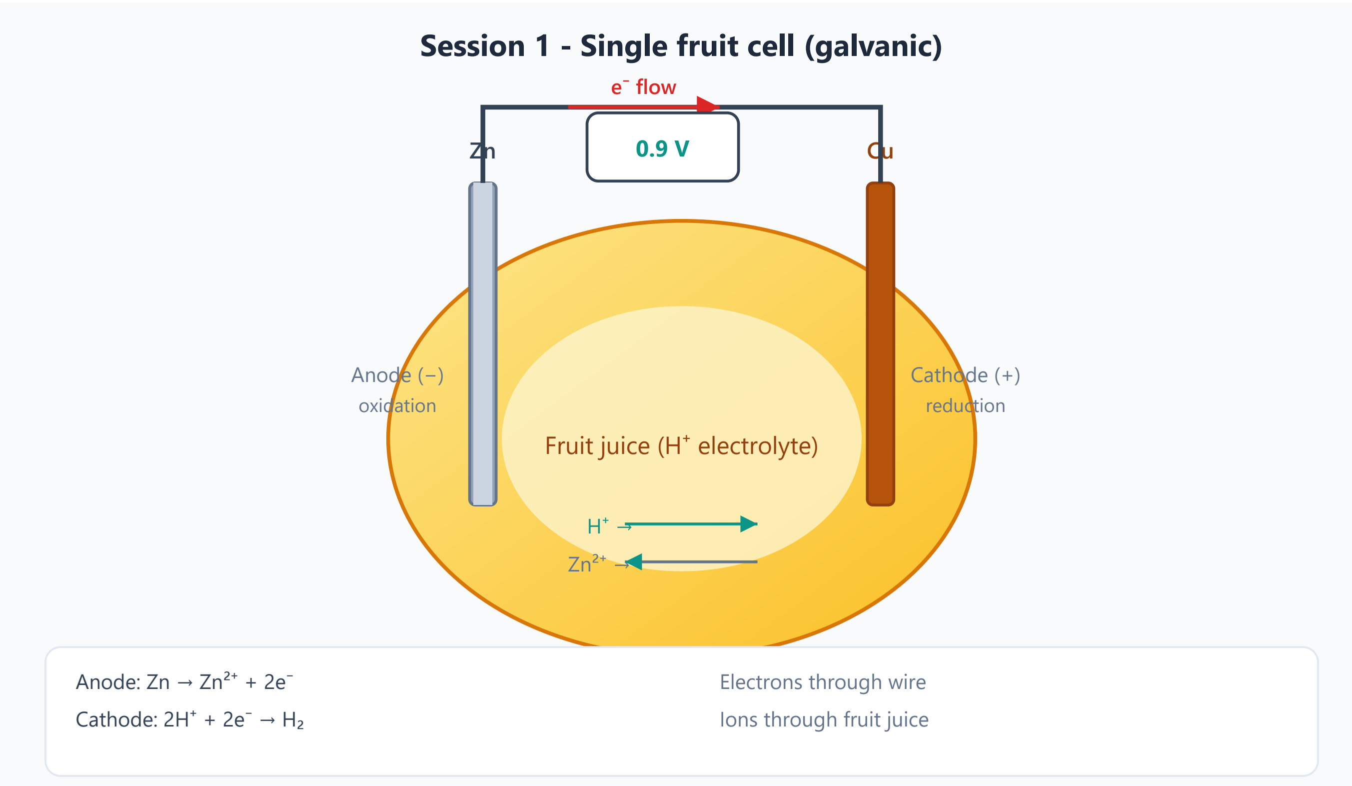

Figure 1 — Single galvanic cell: Zn anode (−), Cu cathode (+), electrons through the wire, ions through the fruit juice.

Materials checklist (per group)

Item	Spec / notes
Fruit	3–6 firm lemons, limes, potatoes, or apples
Zinc electrode	Galvanized nail or zinc strip (2+)
Copper electrode	Copper strip, stiff wire, or clean penny (2+)
Multimeter	DC voltage (and current if available)
Alligator clips	4–6 leads
Load	Red LED (1.8–2.2 V forward) and/or 1.5–3 V buzzer
Optional	pH paper, table salt, aluminum foil

Part 1 — Build one cell (25–45 min)

Procedure

1. Choose one fruit. **Roll** a citrus fruit firmly on the bench for 10–15 s (breaks juice sacs). For potato/apple, squeeze gently or make a shallow slit for better contact.
2. Insert the **zinc** electrode (galvanized nail or zinc strip) about halfway in — do not push all the way through.
3. Insert the **copper** electrode **2–3 cm away** from the zinc. Electrodes must **not touch** inside the fruit (touching shorts the cell → ~0 V).
4. Set the multimeter to **DC voltage** (V⎓), typically the 2 V or 20 V range.
5. Clip **red (VΩmA)** lead to **copper**; clip **black (COM)** to **zinc**.
6. Expected: copper reads **positive** relative to zinc (~0.8–1.0 V for lemon).
7. Record the open-circuit voltage. Wait 2 minutes; record again (note drift).

Expected results

Fruit (typical)	Open-circuit voltage
Lemon / lime	~0.8–1.0 V
Potato / apple	~0.5–0.9 V (varies)

Observations to record

Quantity	Value	Notes
Fruit type		
Electrode materials	Zn: ___ / Cu: ___	
Electrode separation (cm)		
Open-circuit voltage (V)		
Voltage after 2 min (V)		
Meter polarity (which metal +?)		

Troubleshooting

Problem	Likely cause	Fix
~0 V	Electrodes touching	Separate inside the fruit
Very low V	Dry fruit, poor contact	Roll fruit; re-insert; try fresher fruit
Unstable reading	Loose clips	Secure alligator clips on bare metal
Negative voltage	Leads swapped	Swap clips or note polarity and continue

Part 2 — Series battery challenge (45–60 min)

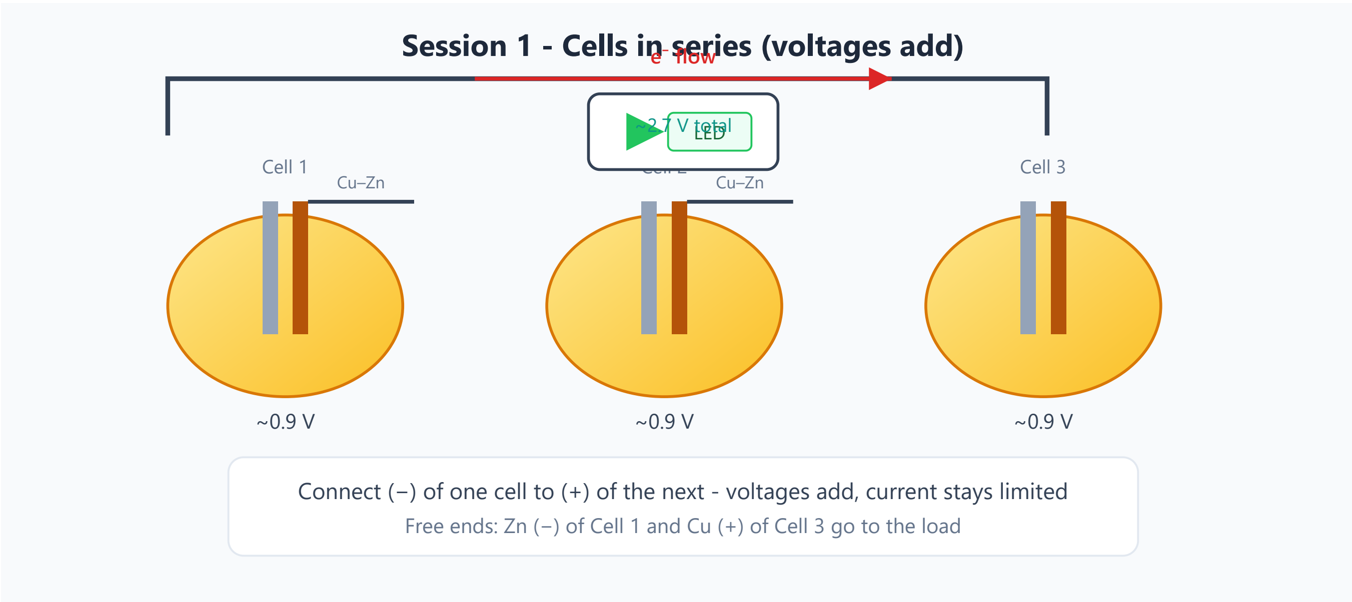

Figure 2 — Cells in series: connect Cu of one cell to Zn of the next; voltages add (~0.9 V each).

Procedure

1. Build **2–3** identical cells (one fruit each) the same way as Part 1.
2. Measure and record each cell's voltage alone first.
3. Wire **in series**:
4. Copper of cell 1 → zinc of cell 2 (alligator lead)
5. Copper of cell 2 → zinc of cell 3
6. Measure total voltage between the **free copper** (positive end) and **free zinc** (negative end).
7. Compare measured total to the sum of the individual voltages.

Challenge question

"If each cell is 0.9 V, how many cells do you need for a 3 V buzzer? For a 1.5 V clock?"

Data table

# cells in series	V_1	V_2	V_3	V_total measured	V_sum individual	Match?
1		—	—			
2			—			
3						

Expected

3 lemon cells ≈ **2.4–3.0 V** open circuit (enough in voltage for many red LEDs; current may still be limiting).

Part 3 — Power something (60–75 min)

Load options (try in order)

1. Multimeter on **DC mA** current range (small load — should work)
2. **Red LED** — longer leg / flat side conventions: longer leg usually **+** (to copper end of series string). Needs ~2 V+ and a few mA. Use **3 cells** first.
3. Active buzzer rated **1.5–3 V DC** (with leads)
4. Low-voltage potato/lemon clock kit (if available)

LED wiring reminder

- Series string: free **copper** → LED **anode (+)**; LED **cathode (−)** → free **zinc**
- If it does not light, reverse the LED once before concluding failure

Measurements

Load	# cells	Voltage under load (V)	Current (mA)	Works?
Open circuit			—	
LED				
Buzzer/clock				

Discussion after attempt

- Did voltage **drop** when connected to a load? (Internal resistance)
- Why might the LED need more **current** than the fruit cell can supply?

Extensions — Side experiments (75–85 min)

Students choose one or more:

A. Compare fruits

Same Zn/Cu electrodes, different fruits → compare open-circuit voltage.

B. Compare metals

Same fruit; replace Zn with aluminum foil, steel nail, or Mg (if available) vs. copper. Record V and note any oxide-film issues (especially Al).

C. pH changes

After 10+ min under load, touch pH paper near each electrode. Hypothesis: cathode region may become less acidic.

D. Salt addition

Add a **pinch** (~0.25 tsp) of table salt into a slit in the fruit; re-measure V and current under the same load.

E. Parallel vs. series

Two cells in parallel (Cu–Cu and Zn–Zn tied): voltage ≈ one cell; compare current capability to series if meters allow.

Extension report template

Variable changed	Control	Result	Explanation (hypothesis)

Wrap-up (85–90 min)

1. Clean stations; fruit goes to waste (not food waste).
2. Quick share: best voltage, whether the LED lit, one surprising result.
3. Preview Session 2: *tomorrow we dissolve copper sulfate and use electricity to plate copper onto a nail.*

Student handout — Quick reference

Build: Roll fruit → Zn + Cu, not touching → measure V
Series: Cu of one cell to Zn of the next → voltages add
LED: Needs ~3 cells + correct polarity; voltage may sag under load
Safety: Do not eat experimental fruit; wash hands after metals

---

Experiment status

- [ ] Procedure pilot-tested with actual fruit/electrodes
- [ ] Typical voltage range documented (record: ___ V)
- [ ] LED/buzzer compatibility verified (# cells needed: ___)
- [ ] Extension options ranked by time available
- [ ] Wrap-up / Session 2 preview timed

SHARED SAFETY NOTES

Syllabus Safety Notes

Apply these across all sessions. Session-specific hazards are listed in each session folder.

General rules

1. **No eating experimental materials.** Fruits, potatoes, and liquids used in experiments are no longer food.
2. **Avoid short circuits.** Batteries can heat up if directly shorted.
3. **Clean surfaces matter.** Electroplating requires clean metal surfaces.
4. **Metal salt waste should be collected.** Copper and silver solutions should not be casually poured down the drain.
5. **Water choice:** tap water is fine for most baths; use **distilled/deionized water for 0.01 M $AgNO_3$** (Session 3) so chloride does not precipitate silver.

Session-specific hazards

Session	Key hazard	Mitigation
1	Sharp nails/wires	Supervise handling; wash hands after metals
2	Copper solutions	Goggles; no ingestion; collect waste
3	Silver nitrate	Gloves + goggles; stains skin/clothing; small volumes
4	Gas evolution + H_2 balloon	No saltwater; ammeter in series; instructor-only balloon/flame demo
5	Corrosion demos + electrolytic derusting	Use washing soda or baking soda only — NOT NaCl; disconnect power when not supervised

Hydrogen balloon / pop demo (Session 4, instructor-only)

- Feed **cathode (H_2) gas only** into a **small** balloon — do not mix with O_2
- Inflate modestly; move away from the electrolysis bath and faces
- Long lighter at arm's length; clear area; extinguisher accessible
- **Never** ignite a sealed rigid container or a mixed H_2/O_2 balloon
- Skip entirely if the venue forbids open flame